Maximum compactness of static stars when the lapse is tied to the isotropic conformal factor, N = 2/ψ − 1

Ugo Lanciano
Independent researcher, Locri, Italy · ORCID 0009-0008-7618-9635

Version 2.34.2, 2026-10-23. Archived on Zenodo, https://doi.org/10.5281/zenodo.23595759. Licence CC BY 4.0. Developed with the help of AI assistants, as declared in the record.

Student version. This PDF is the student version of the record, written with an AI assistant from the archived text. The archived text remains on Zenodo: https://doi.org/10.5281/zenodo.23595759

Abstract

In isotropic coordinates the Schwarzschild metric satisfies N = 2/ψ − 1 for every mass, so if the lapse is a function of the isotropic conformal factor alone, vacuum general relativity fixes that function (in areal coordinates the same argument gives a different closure, Jacobson 2007: the choice of the isotropic variable is an assumption). Imposing this closure inside matter, while keeping the Hamiltonian constraint and energy–momentum conservation of a perfect fluid, gives a phenomenological closure of the static equations in which the fluid pressure does not enter the field equations. Read with Einstein's equations, the stars are anisotropic. For uniform density, in closed form: maximum compactness 2^(1/3) − 2^(−1/3) = 0.4662 against Buchdahl's 4/9, surface redshift 2.847 against 2, and 0.4043 against 3/8 if p_c ≤ ρ is imposed. For five density profiles the maximum compactness is 3–5% above GR. At observed compactness the required central pressure is already lower than in GR (by 7% at C = 0.15), so realistic mass–radius curves differ. With the SLy and APR4 equations of state the closure raises the maximum mass by about 5% and the radius of a 1.4 solar-mass star by 0.1 km, about ten times less than present NICER errors. GR and the closure are both compatible with PSR J0437−4715; at 2.08 solar masses both give radii below the 68% interval of Salmi et al. (2024) for PSR J0740+6620. With current data the closure cannot be distinguished from general relativity.

How compact can a star be? A different rule for the inside of neutron stars

Student version of MM record 2.34.2 (DOI 10.5281/zenodo.23595759), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Ugo Lanciano, 2026. Licence CC BY 4.0.

The versions of this work. This record has had three versions.

- Version 2.34 is the starting one.
- **Version 2.34.1** (October 2026) adds the prior work on the rule, which 2.34 did not cite. The relation N = 2/ψ − 1 inside matter is the assumption already made by Buchdahl in 1964 for his n = 5 solution. Version 2.34.1 also adds two checks: the maximum compactness with "causal" equations of state, and the binding energy of neutron stars. The programs, results and notes of 2.34 stay the same.
- **Version 2.34.2** (October 2026) corrects some figures of 2.34.1. The corrections concern four things: the binding energy at exact mass, the reference value of ALF2, a numerical noise in the radii of the stars, the comparison with NICER. It also adds the checks made for a paper to be sent to a journal: closed forms of the effective stress, a bound C < 12/25 for non-increasing densities, the post-Newtonian expansion of the interior.

The earlier sections are left as they were. **Where they differ from the corrections of 2.34.2, the corrections hold.** In this text, when a number was corrected, we flag it right at the place where it appears.

In short

What a compact star is

A star is held together by its own gravity. The pressure of the matter inside the star pushes outwards and keeps it in equilibrium. A neutron star is a dead star, squeezed very hard. It has roughly the mass of the Sun in a ball as big as a city.

The **compactness** C tells how squeezed a star is. It is computed with the ratio $GM/(Rc^2)$. Here M is the mass, R the radius, G the gravitational constant and c the speed of light. The larger the mass and the smaller the radius, the larger C is.

The Buchdahl limit

In Einstein's general relativity (we write it GR) there is a famous limit. Buchdahl found it in 1959. It holds for a static star, that is, one at rest, made of a perfect fluid, with a density that does not increase going outwards. Such a star cannot be more compact than 4/9. Beyond that value the pressure at the centre would have to be infinite.

A different rule

This work studies a different rule for the inside of a static, spherical star. The rule ties two numbers together.

- The **lapse** N tells the rate of clocks. Near a mass clocks run slower: N is smaller than 1.
- The **isotropic conformal factor** ψ tells how much space there really is in a stretch of the map: near a mass, more rulers are needed to cover the same stretch.

The rule is:

$N = 2/\psi - 1$.

For a single spherical body in vacuum this relation holds exactly. It is a property of the **Schwarzschild metric**, that is, of Einstein's solution for the empty space around a spherical mass, written in isotropic coordinates.

Imposing it inside matter as well changes only one thing in the equations: **pressure no longer acts as a source of the lapse**. In GR pressure contributes to gravity, a bit like mass. Here it no longer does so for the rate of clocks.

Let us read the resulting metric with Einstein's equations. Then it looks as if there were the fluid plus an extra stress, called the **anisotropic effective stress**. Anisotropic means that it is not the same in all directions.

The differences from GR inside matter are small, of order p/ρ. Here p is the pressure and ρ the energy density. In ordinary stars p/ρ is tiny. This is why the differences appear only in neutron stars.

The results

- For uniform density, that is, the same everywhere, the sequence of stars ends at a compactness $2^{(1/3)} - 2^{(-1/3)} \approx 0.466$. It is an exact formula. In GR the limit is instead $4/9 \approx 0.444$. If the density decreases outwards, the maximum is lower (see the table further on).
- For uniform density, at compactness between 0.15 and 0.23, the pressure needed at the centre is 7–12% lower than in GR.
- With two realistic nuclear equations of state the maximum mass rises by about 5%. The radii change by about 0.1 km. This is less than the present errors of the NICER measurements. With these two equations of state GR and the rule are compatible with the star PSR J0437−4715. And both give radii below the 68% interval of Salmi and colleagues for the star PSR J0740+6620.

The rule is an assumption, not a derivation. And it is not yet a theory: there is no action and there is no dynamics. An action is the formula from which all the equations of a theory are obtained. The dynamics tells how things change in time.

Why this rule

In vacuum, for a single body, the lapse depends only on the isotropic conformal factor: $N = 2/\psi - 1$. The Schwarzschild metric depends on the mass only through M/r. So the same formula holds for every mass M.

The assumption of this work is that the same formula also holds inside matter. In GR the Hamiltonian constraint gives ψ from the energy density alone. Pressure, instead, enters the equation for the lapse. Imposing $N = N(\psi)$, that is, N as a function of ψ only, therefore removes pressure as a source of the lapse. The Hamiltonian constraint and the fluid equations stay the same.

The rule was suggested to the author by an informal picture. In this picture the lapse at a place depends only on the conformal factor at that place. It does not depend on the internal stresses of matter. For readers who want it, the picture is explained in a separate document. Nothing in the calculations depends on it.

Neutron stars are the only static objects in which pressure is comparable to the energy density. So they are the place where this rule can be put to the test.

The equations

We use isotropic coordinates and a static, spherical star. The metric is $ds^2 = -N^2dt^2 + \psi^4(dr^2 + r^2d\Omega^2)$. We use units in which $G = c = 1$. ρ is the total energy density measured by static observers. There are three equations.

1. **Hamiltonian constraint, as in general relativity:** $\nabla^2\psi = -2\pi\rho\psi^5$. It tells how the density curves space.
2. **The rule**, in place of GR's lapse equation, which is $\nabla^2(N\psi) = 2\pi N\psi^5(\rho + 6p)$: $N = 2/\psi - 1$, that is, $\psi + N\psi = 2$.
3. **Matter:** a perfect fluid that obeys the conservation of energy and momentum, $\nabla_\mu T^{\wedge}\mu\nu = 0$. That is, $p' = -(\rho + p)N'/N$, with $p = 0$ at the surface. In words: each layer of the star is in equilibrium.

The symbol ∇^2 is called the Laplacian. It measures how much a value at a point differs from the average of the nearby points.

Where the rule comes from, and what is assumed

In isotropic coordinates the Schwarzschild metric has $\psi = 1 + M/2r$ and $N = (1 - M/2r)/(1 + M/2r)$. So $N = 2/\psi - 1$ for every M. Suppose that the lapse is a function of the isotropic conformal factor alone. Then vacuum GR fixes that function in only one way.

The choice of variable matters. One could use areal coordinates. In those, the same argument gives $N^2g_{RR} = 1$ (Jacobson 2007). Inside matter that relation forces $p_r = -\rho$ and allows compactness up to 1/2. The results of this work are a property of the **isotropic** rule. The isotropic variable is the author's choice. It is an assumption: the spatial metric is taken to be conformally flat, with ψ as the physical variable, and the rule is imposed inside matter too. ψ is obtained by integrating the constraint over the whole star. So the rule is a pointwise relation between two fields. It is not a local equation of state.

One body only. All this holds for exact spherical symmetry. Let us try two masses: $\psi = 1 + a_1/|x - x_1| + a_2/|x - x_2|$. This ψ is harmonic. It is the Brill–Lindquist conformal factor for two bodies at rest at one instant. With $N = 2/\psi - 1$ the static metric is not a vacuum solution. Its Ricci tensor, which measures curvature, is not zero. It has a term of order a_1a_2 in an open region. This term vanishes when either of the two masses is removed.

What it implies

The fluid pressure does not enter the field equations. The metric, and hence its effective stress, is fixed by ρ alone.

Let us read the same metric with Einstein's equations. It has an anisotropic effective stress: the radial pressure p_r (outwards) differs from the tangential one p_t (along the surface), with $p_t > p_r$. Already in the Newtonian limit, that is, for weak gravity, the effective radial stress at the centre is half the fluid pressure. The difference between the effective stress and the fluid stress is conserved by itself and has zero energy density.

This is a phenomenological closure of the static equations, not yet a theory: there is no action and there is no dynamics. A phenomenological closure is a practical rule that completes the equations and lets us do the calculations.

Local versus global. Far from the star $\psi \to 1 + M_{ADM}/2r$ and $N \to 1 - M_{Komar}/r$. M_{ADM} and M_{Komar} are two ways of measuring the total mass from far away. Taking the integral over all space one finds $\int\nabla^2(\psi + N\psi)\, d^3x = 4\pi(M_{Komar} - M_{ADM})$. In GR the quantity inside the integral, $2\pi\psi^5[N(\rho + 6p) - \rho]$, is not zero inside matter. But its integral is zero, because the Komar and ADM masses coincide for stationary spacetimes that become flat far away (Beig 1978). The rule, instead, sets that quantity to zero point by point. In GR the balance works out only for the total; here it works out at every point.

Related theories

Other proposals also leave GR unchanged in vacuum (up to a cosmological constant) and change only the interior of matter:

- Eddington-inspired Born–Infeld gravity (Bañados and Ferreira 2010); its neutron stars were studied by Pani, Cardoso and Delsate (2011);
- Palatini f(R) gravity (Olmo 2011).

These theories differ from the rule of this work in two ways. First: they follow from an action, which the rule does not have. Second: their interiors depend on a coupling: κ in Born–Infeld gravity, the function f in Palatini f(R). Here instead, once the isotropic variable is chosen (an assumption, stated above), the relation between N and ψ inside matter is the one fixed by the vacuum solution. The rule has no coupling constant. The equation of state remains an input. For uniform density this gives a single value: 0.466.

Where the differences come from

In these variables ψ is sourced only by the energy density, both in GR and with the rule. This happens through the same Hamiltonian constraint. The only difference is the equation for the lapse.

In GR pressure is a source of the lapse. In covariant form this is written $D^2N = 4\pi N(\rho + 3p)$. In the flat-Laplacian form used above it becomes $\nabla^2(N\psi) = 2\pi N\psi^5(\rho + 6p)$. Here instead the lapse follows from ψ, so pressure is not a source of it.

What does this mean in practice? In GR pressure adds gravitational pull. More pressure, more gravity, and even more pressure is needed. With the rule pressure is no longer a source of the lapse, and there is less gravitational pull due to pressure. Then the same star needs less pressure at the centre to stay in equilibrium. And a given equation of state supports a little more mass.

The two sequences of stars end for different reasons. In GR the sequence of uniform-density stars ends at 4/9, where the central pressure needed becomes infinite. With the rule it ends where the lapse at the centre reaches zero, ψ_c = 2, at 0.466. There too the pressure becomes infinite.

The effect is of relative order p/ρ. It is present in any matter. But in the Sun or the Earth p/ρ is tiny, and GR and the rule agree to that accuracy. Only in neutron stars does p/ρ reach tens of per cent. Only there do the two separate measurably.

Where the sequence of stars ends

Imagine building more and more compact stars, one after the other. Sooner or later the sequence ends: beyond a certain point the star cannot stay in equilibrium.

For uniform density one thing holds, both in GR and with the rule: the product (ρ + p)N is constant. So the pressure at the centre becomes infinite exactly when the lapse at the centre reaches zero. In GR pressure is a source of the lapse (ρ + 6p in the flat-Laplacian form). So pressure comes back and acts on the geometry. With the rule, instead, N is fixed by ψ, and ψ depends only on ρ. The sequence ends at ψ_c = 2. It is the edge of the range of values where $N = 2/\psi - 1 \geq 0$. At ψ_c = 2 the lapse at the centre is zero: there the clocks stop.

Uniform density, with an exact formula. The constraint is the Lane–Emden equation with n = 5. Its solution is $\psi = \psi_c(1 + 2\pi\rho\psi_c^4 r^2/3)^{(-1/2)}$. Here ψ_c is the value at the centre. Matching to the exterior gives $\psi_R = \psi_c^{(2/3)}$. The compactness is $C = GM/(Rc^2) = 2(\psi_R - 1)/\psi_R^2$, with R the areal radius.

- GR's Buchdahl limit, C = 4/9, corresponds to ψ_R = 3/2 (ψ_c = 1.8371).
- The limit ψ_c = 2 gives **C_max = 2^(1/3) − 2^(−1/3) = 0.46622**. It can also be said this way: the redshift of the light leaving the surface is z = 2.847 (in GR it is 2). The redshift tells how much light is stretched as it climbs out of the star's gravity. This value is below Ivanov's bounds for anisotropic stars: 3.842 or 5.211, depending on the energy condition imposed on the tangential pressure.
- Let us require the fluid pressure to satisfy p_c ≤ ρ. Then one finds the equation $y^3 + 2y - 4 = 0$, with $y = \psi_c^{(1/3)}$. That is, C = 0.40427 (in GR 3/8). This holds for uniform density only.
- The effective radial stress at the centre is $p_r/\rho = (\psi_c - 1)/(3(2 - \psi_c))$.

The link with Buchdahl's hypotheses. There is a theorem by Andréasson (2008). It holds for static, spherical GR solutions with two conditions: p_r ≥ 0 and p_r + 2p_t ≤ ρ. Then 2m/r ≤ 8/9, with no requirement on how the density varies. Here p_r ≥ 0. So to exceed 4/9 one must violate the trace condition, p_r + 2p_t ≤ ρ. This happens at the centre for

$\psi_c > 3/2$, that is, $C > 0.3615$. GR's own uniform star also violates the same condition, for $C > 5/18$. What takes the star outside Buchdahl's hypotheses is the anisotropy. The dominant energy condition for the effective stress fails for $\psi_c > 7/4$ ($C > 0.4289$). But this is not needed to exceed 4/9.

Other density profiles (computed numerically). Take stars where the density decreases from the centre outwards following a shape f(x): $\rho(r)/\rho_c = f(x)$, with $x = r/R$. Careful: here r is the **isotropic** radius. This is why the values cannot be compared with the published ones for profiles of the same name, which use the areal radius. Compare only within a row. The last digit is ±0.0005.

f(x)	maximum compactness, isotropic rule	GR, same profile and same coordinate	ratio
1	0.4662	0.4444	1.049
$1 - x^2$	0.4253	0.4052	1.049
$1 - x$	0.4084	0.3922	1.041
$(1 - x^2)^2$	0.3937	0.3750	1.050
$(1 - x)^2$	0.3573	0.3457	1.034

For every profile tested the maximum compactness is 3–5% higher than in GR. Among these profiles the most compact is the uniform one. This, however, is not proved for all profiles that do not increase outwards. (Version 2.34.2 adds a proved bound, 12/25: see further on.)

Neutron stars below the limit (uniform density)

Take the same profile and the same ψ_c. Then the geometry of space is the same as in GR. So M and R are the same too. The lapse and the pressure, instead, are not. For uniform density the pressure needed at the centre is lower than in GR:

C	0.15	0.23	0.29	0.34	0.375
p_c (rule) / p_c (GR)	0.93	0.88	0.82	0.75	0.67

For example, at $C = 0.29$ the rule needs 82% of GR's central pressure.

Realistic equations of state and NICER

What an equation of state is. An **equation of state** tells how much pressure matter gives at a certain density. Think of a sponge and a stone. If you squeeze them, the sponge gives way at once, the stone resists. The nuclear matter of neutron stars has its own "hardness", still not well known. Scientists use several possible equations of state: SLy, APR4 and others. A "stiffer" one resists more.

How it enters the calculations. The same three equations as above are used. But with a realistic equation of state $\varepsilon(p)$ for cold nuclear matter. Here ε is the total energy density, which above we called ρ. So:

- ψ comes from the Hamiltonian constraint $\nabla^2\psi = -2\pi\varepsilon\psi^5$ (ε the total energy density);
- $N = 2/\psi - 1$;
- p comes from conservation: $p' = -(\varepsilon + p)N'/N$.

The central value ψ_c is fixed by requiring $\psi \to 1$ at infinity. The mass is read from the exterior, $\psi = 1 + M/2r$. The areal radius is $\psi(R)^2 R$. The comparison is with GR, that is, with the TOV equations, using the same equation of state. We use the piecewise-polytrope forms of SLy and APR4 of Read and colleagues (2009), with their SLy crust. Stars are counted on the branch with $dM/dp_c > 0$, that is, where the mass grows with the central pressure. For the rule this static criterion is an assumption, because there is no dynamics.

Six predictions registered before the calculation. For this comparison six predictions were written down, fixed with a SHA-256 fingerprint before the calculation:

1. the GR values reproduce Read and colleagues within 2%;
2. the rule raises the maximum mass by at least 3%;
3. it shifts the radius at 1.4 M☉ by less than 0.5 km,
4. upwards;
5. GR and the rule both fall in the interval of PSR J0437−4715;
6. both reach at least 2.01 M☉.

All six held. This section has not yet been re-run independently. The symbol M☉ stands for the mass of the Sun: 1.4 M☉ means 1.4 times the mass of the Sun.

EOS	theory	M_max (M☉)	R at 1.4 M☉	R at 1.418 M☉	R at 2.08 M☉
SLy	GR	2.048	11.71 km	11.69 km	— (not reached)
SLy	rule	2.146	11.84 km	11.83 km	10.83 km
APR4	GR	2.188	11.33 km	11.33 km	10.68 km
APR4	rule	2.298	11.43 km	11.42 km	11.03 km

Corrected in version 2.34.2 (corrections 3 and 4): the radii of the rule had a small numerical noise, and the right mass for PSR J0740+6620 is 2.073 M☉. The updated table is further on.

The GR values agree with Read and colleagues within 1.1%. Their piecewise forms differ from the tabulated equations of state by the same amount.

The comparison with NICER (68% intervals). NICER is an instrument on the International Space Station. It measures the mass and radius of some rotating neutron stars, the **pulsars** (the PSR labels).

- PSR J0437−4715: M = 1.418 ± 0.037, R = 11.36 (+0.95/−0.63) km (Choudhury and colleagues 2024). All four curves fall inside.
- PSR J0740+6620: M = 2.08 ± 0.07, R = 12.49 (+1.28/−0.88) km (Salmi and colleagues 2024). Dittmann and colleagues 2024 give 12.92 (+2.09/−1.13) km. GR and the rule, with these two equations of state, give radii at 2.08 M☉ below the interval of Salmi and colleagues. The rule is closer, by about 0.35 km (APR4).

What this means. With the same equation of state the rule raises the maximum mass by about 5%. It moves the radius of a 1.4 M☉ star by 0.10–0.13 km. *(Corrected in 2.34.2, correction 3: 0.12 km for SLy and 0.10 km for APR4.)* This shift is about ten times smaller than the present NICER errors, which are about ±0.8 km. So **today's data do not distinguish the rule from GR**.

The 5% in the maximum mass is mixed up with the stiffness of the equation of state. Version 2.34.1 wrote: "a slightly stiffer equation of state in GR gives the same". ***This sentence is withdrawn in version 2.34.2 (correction 7): it had not been computed.*** It remains true that the increase is as large as the differences between the equations of state.

To separate the two, radius errors of about 0.1 km would be needed, or an independent determination of the equation of state. With SLy and APR4 the rule is compatible with PSR J0437−4715. Like GR, it gives radii below the interval of Salmi and colleagues for PSR J0740+6620. These data do not confirm it.

What is not claimed

- It is not claimed that the rule holds inside matter: it is assumed.
- It is not claimed that the isotropic variable is to be preferred over the others: this too is assumed.
- There is no covariant or dynamical formulation. So nothing is said on radial stability (whether the star holds up under small pushes), on rotation or on the post-Newtonian constraints of a dynamical completion. *(Version 2.34.2 adds two post-Newtonian readings: see further on.)*
- No bound is claimed over all density profiles or over all equations of state. *(Version 2.34.2 proves a bound, 12/25, for profiles that do not increase outwards.)*
- Causality is not claimed. For uniform density the speed of sound is infinite both in GR and with this rule. In physics nothing should go faster than light.
- It is not claimed that NICER data favour the rule over GR. With SLy and APR4 the two cannot be told apart within today's errors.

Added in version 2.34.1

Who had already done it: Buchdahl's 1964 assumption

The relation N = 2/ψ − 1 inside matter is not new.

- **Buchdahl's choice.** In 1964 Buchdahl was looking for a relativistic version of the Emden polytrope of index 5. A polytrope is a simple model of a star. Buchdahl chose isotropic coordinates with ψ = 1 + f and N = (1 − f)/(1 + f) (his equation 2.2). That is exactly N = 2/ψ − 1. His reason was the same one given above: the exterior Schwarzschild metric has this form (his equation 2.3).
- **His calculation.** Then he required the matter to be a perfect fluid. The isotropy of the pressure, that is, pressure equal in all directions, gives f f″ − 3f′² − f f′/r = 0 (his equation 2.7). The solution is f = (A + Br²)^(−1/2) (his equation 2.8).
- With A = 0 one recovers the Schwarzschild exterior.
- With A > 0 one has his n = 5 solution. It has infinite radius and finite mass.

A program of the package redoes this calculation. Version 2.34 cited Buchdahl (1959) for the 4/9 limit, but not this paper.

What this work does differently. Buchdahl requires the metric to be a solution of Einstein's equations with a perfect fluid. This leaves only his n = 5 family. Here instead the relation takes the place of the lapse equation, for any density profile:

- the matter is still a perfect fluid, with the pressure following from conservation;
- the stars have a finite radius;
- what is dropped is the isotropy of the effective stress, the one obtained by reading the metric with Einstein's equations: that effective stress is anisotropic.

The exact maximum 2^(1/3) − 2^(−1/3) for uniform density, the five profiles and the realistic equations of state are consequences of this second choice.

Other prior work. It was found by a novelty check made on 7 October 2026:

- the metric of Buchdahl's solution in the form ψ = 1 + f is the entry "Buch2" of the catalogue of Delgaty and Lake (1998);
- the conformal factor of the uniform-density star is the n = 5 Lane–Emden solution of the Hamiltonian constraint. It was studied as initial data, without a lapse, by Baumgarte, Ó Murchadha and Pfeiffer (2007);
- Puttasiddappa, Bretón and Pérez Bergliaffa (2026) build uniform-density anisotropic stars, also beyond the Buchdahl limit. In their stars the anisotropy is proportional to ρ k^μ∇_μP_r with a constant coefficient. The stars of this work are not in that family: the corresponding ratio changes along the radius.

Causal equations of state and the turning point

An equation of state is **causal** if the speed of sound in matter does not exceed the speed of light. As a formula: dp/dε ≤ 1. The predictions of this part were registered before the calculation.

- **Equation of state used:** p = c_s²(ε − ε_s). Here c_s² is the square of the speed of sound. c_s² = 1 is the stiffest of the causal ones: sound travels at the speed of light.
- **Stability:** assessed only by the **turning point**, that is, by the maximum of the mass along the sequence. For barotropic stars in GR this criterion is exact. The rule has no dynamics, so for it the criterion is only an indicator.

c_s²	compactness at maximum mass, GR	compactness at maximum mass, rule	maximum mass, rule/GR
1	0.3527	0.3663	+6.2 %
1/2	0.3040	0.3174	+5.6 %
1/3	0.2691	0.2796	+5.0 %

Corrected in version 2.34.2 (correction 6): the maximum was later located precisely, not read on the grid. The updated table is further on.

- **Program check.** The GR value 0.3527 agrees with the 0.354 of the literature (Koranda, Stergioulas and Friedman 1997), within the resolution of the grid of central pressures. *(Version 2.34.2 specifies to whom this number should be attributed: correction 6.)*
- **Compactness.** With the stiffest causal equation of state the rule reaches 0.366 at maximum mass. This is below the 0.389 of Alho, Natário, Pani and Raposo (2022) for configurations that are causal and also radially stable. And it is below their 0.462 for configurations that are only causal. The value 0.466 remains the one of incompressible matter, to be compared with Buchdahl's 4/9.
- **Energy conditions.** Let us read with Einstein's equations the maximum-mass rule star with $c_s^2 = 1$. It satisfies the dominant energy condition everywhere.
- **Outcome.** Five registered predictions out of five held.
- **Limits.** Here causality refers to the fluid ($dp/d\varepsilon \leq 1$). The effective medium read in GR has no defined wave speeds, without a model of the anisotropy. And a true analysis of radial stability would need a dynamics.

The binding energy of neutron stars

The **binding energy** tells how much energy is released when the star forms. Think of a brick wall: you weigh all the loose bricks, then the finished wall. For a real wall the difference cannot be seen. In a neutron star, instead, the whole star weighs less than its matter counted piece by piece. The difference, $BE = M_B - M$, is the binding energy. M_B is the baryonic mass, that is, the mass of all the matter counted piece by piece. M is the gravitational mass, the one measured from outside.

The predictions of this part were registered before the calculation. A check program was written after the calculation and is not registered.

- **Equations of state.** The 17 piecewise polytropes of Read and colleagues (2009) are used. Their parameters had been typed from memory. For this reason they were checked against the published maximum masses: all 17 agree within 1.1 %. *(Corrected in 2.34.2, correction 2: for ALF2 the reference value was wrong, and the admitted equations of state are 16, not 17.)*

Results.

- **At fixed mass the binding energy is nearly the same in the two theories.**
- Between 1.25 and 1.6 $M_\odot$ the rule equals GR within 0.1 %. *(Corrected in 2.34.2, correction 1: it becomes "lower by less than 0.2 %".)*
- At the GR maximum mass it is lower by 0.4–0.75 %, even though the radius there grows by 0.61–1.18 km. *(Corrected in 2.34.2, correction 1: 0.43–0.80 % and 0.74–1.36 km.)*
- A likely reason is that both theories have the same Hamiltonian constraint. It is the constraint that fixes the geometry of space from the energy density. This is a reading of the numbers, not a proof.
- **The quasi-universal relation BE/M(β) does not distinguish the rule.** Here β is the compactness. The mean offset is +2.4 %, while the scatter of GR is 4.3 %.
- **The pulsar PSR J0737−3039B** (Podsiadlowski and colleagues 2005) selects the same five equations of state in the two theories.

Two predictions did not hold. Here is the registered text, translated from Italian, and then the outcome.

- **E2** (a check in GR): "the fit $BE/M = d_1\beta + d_2\beta^2$ over all stable stars of the admitted equations of state has d_1 between 0.55 and 0.70 and a relative (rms) scatter below 5 %". Outcome: $d_1 = 0.480$ and $d_2 = 0.634$, with scatter 4.34 % (505 stars). The scatter holds, d_1 does not.
- **E3** (same equation of state, same mass $M = 1.2489\ M_\odot$): "the closure gives a lower binding energy than GR, between 0.5 % and 5 % less, for every admitted equation of state". Outcome: the differences lie between −0.10 % and +0.02 %. *(Corrected in 2.34.2, correction 1: with the exact mass the rule is lower by 0.02–0.05 %; E3 still does not hold.)*

A caveat on the absolute values. The registered range for d_1 came from published fits (Lattimer and Prakash 2001; Breu and Rezzolla 2016). So the outcome of E2 means that our GR binding energies are lower than those fits. A possible cause is the unit of baryon mass. It is not isolated here. This does not affect the comparison between the two theories, which use the same definition. It does affect, instead, which equations of state are compatible with J0737−3039B.

Three registered predictions out of five held.

What changes in what is claimed (version 2.34.1)

- **The relation is Buchdahl's.** This work proposes to use it as a rule for any density, and studies its consequences.
- **Causal equations of state.** Using the turning point as an indicator, the rule reaches C = 0.366 at maximum mass. This is below the cited benchmarks: 0.389 (causal and radially stable configurations) and 0.462 (causal configurations). The radial stability of the rule has not been established.
- **With today's uncertainties on the equation of state, the static observables tested do not separate the two theories.**
- The maximum mass rises by +3.7 to +5.3 % for the 17 realistic equations of state, and by +5.0 to +6.2 % for the causal ones. This increase, and the radius near the maximum mass, are mixed up with the equation of state.
- The binding energy does not separate them either.
- More precise mass and radius measurements (radii to about 0.1 km) or an independently constrained equation of state could help. Tidal deformability and the oscillations of the star would also be useful. But the rule has no dynamics to compute them.

Added in version 2.34.2

These are the checks made while preparing a journal paper on this rule. They correct some figures of version 2.34.1. The earlier sections are left as they were. **Where they differ from this section, this section holds.**

The corrections

1. **Binding energy at fixed mass.** Version 2.34.1 obtained the binding energy by interpolating on a grid of stars. Now it is computed at the exact gravitational mass. The value of the rule is lower than the GR value for every admitted equation of state:
2. by 0.02–0.05 % at 1.2489 M☉, and by 0.03–0.17 % at 1.6 M☉ (in version 2.34.1: −0.10 to +0.02 % and −0.15 to +0.02 %);
3. at the GR maximum mass, by 0.43–0.80 %, with radii larger by 0.74–1.36 km (in version 2.34.1: 0.4–0.75 % and 0.61–1.18 km).

The registered prediction E3 (0.5 to 5 % lower at 1.2489 M☉) still does not hold. The sentence "the closure equals GR within 0.1 % between 1.25 and 1.6 M☉" becomes "lower by less than 0.2 %".

1. **ALF2.** The binding-energy program used 1.991 M☉ as the reference maximum mass of the equation of state ALF2. Table III of Read and colleagues (2009) instead gives 2.086 M☉. The registered admission rule requires the GR maximum mass to be within 2 % of the tabulated one. With the correct value, ALF2 is excluded. Our GR value, 1.976 M☉, reproduces their piecewise fit, which in turn differs by −5.3 % from the tabulated equation of state.
2. The admitted equations of state are 16, not 17.
3. Without ALF2 these do not change: the range of the maximum-mass increase (+3.7 to +5.3 %); the mean offset of the rule from the GR fit BE/M(β) (+2.4 %, against a scatter of 4.4 % about our GR fit); the five equations of state selected by PSR J0737−3039B in the two theories.
4. The original program and its results are left as deposited.
5. **The radii of the rule stars.** With the settings of the first program (relative tolerance 10^{-9}, maximum step 0.05 km) the radius of the rule stars has a numerical noise of about 0.01 km, up to 0.03 km. It comes from the crust of the star. With tight settings (10^{-11}, maximum step 5 m) the curves are smooth to about 0.001 km. The masses do not change.

EOS	theory	M_max (M☉)	R at 1.4 M☉	R at 1.418 M☉	R at 2.073 M☉
SLy	GR	2.0485	11.709 km	11.698 km	— (not reached)
SLy	rule	2.1456	11.831 km	11.823 km	10.864 km
APR4	GR	2.1881	11.321 km	11.317 km	10.700 km
APR4	rule	2.2980	11.422 km	11.420 km	11.039 km

The radius shift at 1.4 M☉ is 0.12 km (SLy) and 0.10 km (APR4), not 0.13 and 0.10 km.

1. **NICER.**
2. The mass 1.418 ± 0.037 M☉ for PSR J0437−4715 is the final (posterior) estimate of Choudhury and colleagues

(2024).

3. For PSR J0740+6620, Salmi and colleagues (2024) give 2.073 ± 0.069 M⊙. The value 2.08 ± 0.07 M⊙ is the mass measured with radio signals, which they use as a starting point (prior). The conclusions do not change: the radii lie below their interval, and SLy in GR does not reach this mass.
4. A later analysis of PSR J0437−4715 includes a modulated non-thermal component (Miller and colleagues 2026). It gives 11.9–15.5 km at 68 %. All four curves lie below it: 11.32–11.82 km at 1.418 M⊙.
5. **APR4 and MPA1 are not causal near the maximum mass.** At the centre of the maximum-mass stars the piecewise fit has dp/dε = 1.38 (GR) and 1.31 (rule) for APR4, and 1.09 and 1.04 for MPA1. That is, sound would travel faster than light. For SLy the GR value there is 1.00, right at the causal limit. The upper part of these sequences lies outside the causal range of the fit.
6. **Causal equations of state.** The maximum of the mass is now located precisely, not read on the grid:

c_s^2	compactness at maximum mass, GR	compactness at maximum mass, rule	maximum mass, rule/GR
1	0.3540	0.3668	+6.2 %
1/2	0.3060	0.3157	+5.6 %
1/3	0.2707	0.2784	+5.0 %

- The GR value for $c_s^2 = 1$ agrees with the 0.354 of Lattimer and Prakash (2016).
- The conjecture that this equation of state gives the most compact stars is stated in their formulation, based on Koranda, Stergioulas and Friedman (1997). Version 2.34.1 attributed the number to the latter.
- **A sentence withdrawn.** Version 2.34.1 wrote that "a slightly stiffer equation of state in GR gives the same" maximum mass as the rule. This had not been computed. What remains is only that the increase is as large as the differences between the equations of state.

The new results

The effective stress with exact formulas. Let us read with Einstein's equations any configuration of the rule, with any density profile. The symbolic calculation gives:

- $\rho_{eff} = \rho$;
- the trace $S = p_r + 2p_t = \rho(\psi - 1)/(2 - \psi)$;
- $p_r = -\psi'[(r\psi)' - 1]/[2\pi r\psi^5(2 - \psi)] \geq 0$.

The trace condition, $p_r + 2p_t \leq \rho$, holds exactly where $\psi \leq 3/2$.

A bound for densities that do not increase. Every configuration of the rule with a density that does not increase outwards has **$C < 12/25 = 0.48$**. The proof has three steps:

1. $q = \rho\psi^5$ does not increase; so $\tilde{m}(r)/r^3$, with $\tilde{m} = -r^2\psi'$, does not increase;
2. it follows that $\psi_c - \psi_R \geq (\psi_R - 1)/2$;
3. the lapse at the centre must be positive, $\psi_c < 2$; this gives $\psi_R < 5/3$.

Some notes:

- **Initial data.** The bound uses only the Hamiltonian constraint and $\psi_c < 2$.
- **Without the non-increase condition** thin shells approach $C = 1/2$, that is, $2M/R \to 1$.
- **Equation of state.** For stars built from a barotropic equation of state the density automatically does not increase.
- **Numerical checks.** No step profile exceeds the uniform value 2^(1/3) − 2^(−1/3) = 0.46622. This value remains a **conjecture** for the true maximum:
- out of 120 non-increasing step profiles taken at random, the largest is 0.4662163;
- a local optimisation over two-step profiles finds the uniform one again. The program was written independently by an AI assistant acting as a referee.

The interior with the post-Newtonian expansion. The post-Newtonian expansion is a way of computing, step by step, the corrections to Newton's gravity. To post-Newtonian order the spatial metric is the same in the two theories. The lapse, instead, gives:

- $-g_{00} = 1 - 2U + 2U^2 - 4\Phi_2 - 6\Phi_4$ in GR;

- $-g_{00} = 1 - 2U + 2U^2 - 5\Phi_2$ with the rule.

Here U, Φ_2, Φ_4 are "potentials", that is, numbers that depend on how matter is distributed. The second line can be read in two ways.

- **Read within GR.** The metric is that of GR with an effective stress whose trace is S = ρU/2 to leading order. All the post-Newtonian parameters are those of GR. The rule becomes a statement about matter. But that matter is **nonlocal**: through ψ, the stress at a point depends on the density of the whole configuration, inside and outside r, and on the normalization of ψ at infinity. This is a stronger nonlocality than in the equations of state of Hernández and Núñez (2004), which depend only on the matter enclosed within r.
- **With p as the physical pressure.** Then the rule acts as a modification of gravity inside matter. Some parameters are found:
- ξ = 0 is forced. Outside a sphere, Φ_W has a term in $1/r^4$ that no other potential cancels, and the exterior of the rule is exactly Schwarzschild. There is a Monte Carlo check, that is, one with random numbers.
- Matching the remaining coefficients gives $\beta = \gamma = 1$, $\zeta_3 = 0$, $\zeta_2 = 1/2$ and $\zeta_4 = -1$.
- Every theory derived from an invariant action has $\alpha_3 = \zeta_i = 0$ (Lee, Lightman and Ni 1974).
- Binary pulsars bound $|\alpha_3 + \zeta_2| < 4\times10^{-5}$, and hence ζ_2 (Will 2014). In strong field $|\zeta_2| < 1.3\times10^{-5}$ (Freire and Wex 2024).
- **So, in this reading, the rule is not the static limit of any metric theory derived from an action, nor of any metric theory compatible with these bounds.** It is a reading that falls: the rule taken as modified gravity.
- Moreover, by an argument of Will (1976, 2014), in any reasonable theory $6\zeta_4 = 3\alpha_3 + 2\zeta_1 - 3\zeta_3$. Then $\zeta_4 = -1$ with $\zeta_3 = 0$ would require $3\alpha_3 + 2\zeta_1 = -6$, while $|\alpha_3| < 4\times10^{-20}$.

Causal stars: the effective stress.

- **Slopes.** In the maximum-mass rule star with $c_s^2 = 1$, the slopes $dp_r/d\varepsilon$ and $dp_t/d\varepsilon$ of the effective stress do not exceed 0.922 and 0.917. They are not characteristic speeds, that is, the speeds at which waves would travel in the medium: the effective medium has no prescribed response to perturbations.
- **Dominant energy condition.** It holds in this star.
- **The trace at the centre of the maximum-mass stars.** Let us compare the rule's S/ε with GR's $3p_c/\varepsilon_c$:
- $c_s^2 = 1$: 1.52 against 2.01;
- $c_s^2 = 1/2$: 0.90 against 1.13;
- $c_s^2 = 1/3$: 0.65 against 0.79;
- SLy: 1.20 against 1.62;
- APR4: 1.52 against 2.11.

What changes in what is claimed (version 2.34.2)

- **Density profiles.** In "What is not claimed" there was "a bound over all density profiles". Now there is a proved bound, 12/25, for profiles that do not increase. The maximum 0.4662 is a conjecture.
- **Post-Newtonian constraints.** "Nothing on … post-Newtonian constraints" is replaced by the two readings above.
- **Everything else** stays as in version 2.34.1.

How the work was done

Assistants and review. The calculations were done with the help of AI assistants. The results that come before the section on realistic equations of state were re-run by a second AI assistant, not by a human referee (`documenti/VERIFICA_ESTRATTO.md`). The working notes, in Italian, are in `documenti/`. The author's motivation for the rule is explained in `WHERE_THIS_COMES_FROM.md`. A fuller account of the author's model is in preparation.

Registered predictions. The predictions were written in a file and fixed with a SHA-256 fingerprint before the calculation:

- realistic equations of state and NICER, six predictions: `documenti/PREVISIONI_MASSA_RAGGIO.md`;
- causal equations of state: `documenti/causalita_stabilita/PREVISIONI.md` (SHA-256 826bfd73…, 7 October 2026);
- binding energy: `documenti/energia_legame/PREVISIONI_ENERGIA_LEGAME.md` (SHA-256 80a36072…, 10 October 2026).

The novelty check of 7 October 2026 is in `documenti/novita/NOVITA_ESITO.md`, in Italian.

Where the programs cited in the text are.

- Two centres and Komar–ADM flux: `codice/domini_e_masse.py`.
- Buchdahl's 1964 calculation: `codice/buchdahl_1964_isotropia.py`.
- Comparison with Puttasiddappa and colleagues: `codice/confronto_put.py`.
- Causal equations of state: `codice/causale.py`, output `risultati/causale.log`; in 2.34.2 `codice/causale_massimo.py` and `codice/sforzo_causale.py`.
- Binding energy: `codice/energia_legame.py` and `codice/ricontrollo_masse.py` (the latter written after the calculation, not registered); outputs in `risultati/`; in 2.34.2 `codice/legame_massa_esatta.py`, `codice/legame_massa_massima.py`, `codice/energia_legame_senza_ALF2.py`.
- Mass–radius curves: `codice/massa_raggio.py`, figure `risultati/massa_raggio.png`; with tight tolerances `codice/massa_raggio_stretto.py`, figure `risultati/fig3_mass_radius.pdf`; radius noise `codice/rumore_raggio.py`.
- Speed of sound at the centre: `codice/velocita_centro.py`.
- Symbolic effective stress: `codice/traccia_sforzo.py`.
- Step profiles: `codice/profili_a_gradini.py`, `codice/gradini_ottimizzazione.py`.
- Post-Newtonian interior: `codice/ppn_interno.py`, `codice/phi_w_sfera.py` (the Monte Carlo check is `phi_w_sfera.py`).

How to redo the calculations. Each table is reproduced by a program in `codice/`, with Python 3 and the libraries numpy, scipy, sympy:

- `sforzo_efficace_342.py`: closed forms, pressures, effective stress;
- `stelle_340.py`: five profiles in the same coordinate;
- `stella_334.py`, `densita_338.py`, `pressione_341.py`: supporting checks;
- `einstein_vuoto_esatto.py`: vacuum;
- `domini_e_masse.py`: two centres, and the Komar–ADM flux;
- `massa_raggio.py`: mass–radius curves with SLy and APR4, GR and rule, and the figure (a few minutes).

Added in version 2.34.1 (run them from `codice/`; the JSON files are written in the current folder):

- `buchdahl_1964_isotropia.py`: Buchdahl's calculation, symbolic and numerical (a few minutes);
- `confronto_put.py`: comparison with the anisotropy law of Puttasiddappa and colleagues (2026);
- `causale.py`: causal equations of state, turning point, energy conditions;
- `energia_legame.py 2` (2 = number of processes), then `ricontrollo_masse.py`: binding energy, 17 equations of state.

Added in version 2.34.2 (run them from `codice/`; the JSON files are written in the current folder; the programs that use the output of another one read it from the current folder or from `risultati/`):

- `causale_massimo.py`, then `sforzo_causale.py`: causal equations of state, maximum located precisely, effective stress;
- `massa_raggio_stretto.py` (about half an hour with 2 processes), then `figure_articolo.py <folder>`: SLy and APR4 with tight tolerances, and the figures of the paper;
- `rumore_raggio.py`: the numerical noise of the radius with the settings of `massa_raggio.py`;
- `legame_massa_esatta.py 1` and `legame_massa_massima.py 1` (1 = number of processes), then `velocita_centro.py`: binding energy at exact mass for the 16 admitted equations of state, and dp/dε at the centre;
- `energia_legame_senza_ALF2.py`: the binding-energy relations of version 2.34.1 without ALF2;
- `traccia_sforzo.py`, `ppn_interno.py`, `phi_w_sfera.py`: effective stress (symbolic), post-Newtonian expansion, the potential Φ_W;
- `profili_a_gradini.py 2` and `gradini_ottimizzazione.py`: step profiles.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a complete Italian translation was made, then the rewriting for a school student. The archived text calls the rule a *closure*; here we mostly call it "rule". The numbers were checked with a program against the archived text. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **Action.** The formula from which all the equations of a theory are obtained. The rule of this work does not have one.

- **Anisotropic.** Different depending on the direction. An anisotropic stress pushes differently outwards and along the surface.
- **Binding energy.** The difference between the mass of the matter counted piece by piece and the mass of the whole star.
- **Buchdahl limit.** In GR, a perfect-fluid star with non-increasing density cannot exceed $C = 4/9$.
- **Causal.** An equation of state is causal if sound in matter does not go faster than light: $dp/d\varepsilon \leq 1$.
- **Closure, rule.** The relation $N = 2/\psi - 1$ imposed inside matter as well, in place of GR's lapse equation.
- **Compactness C.** How squeezed a star is: $C = GM/(Rc^2)$.
- **Conformal factor ψ.** The number that tells how much real space there is in a stretch of the map, at a place, compared with a place very far away: near a mass more rulers are needed for the same stretch.
- **Equation of state.** The rule that tells how much pressure matter gives at a certain density, that is, how "hard" it is.
- **General relativity (GR).** Einstein's theory of gravity.
- **Lapse N.** The number that tells how slowly a clock runs at a place, compared with a distant clock.
- **M⊙.** The mass of the Sun, used as a unit.
- **Neutron star.** A dead star, squeezed very hard: about the mass of the Sun in a ball as big as a city.
- **NICER.** An instrument on the International Space Station that measures the mass and radius of pulsars.
- **Post-Newtonian.** The way of computing, step by step, the corrections to Newton's gravity.
- **Pulsar (PSR).** A rotating neutron star that sends out regular pulses.
- **Turning point.** The maximum of the mass along a sequence of stars. In GR, beyond that point the stars are not stable; for the rule, which has no dynamics, it is only an indicator.

References

- H. A. Buchdahl, Phys. Rev. 116, 1027 (1959).
- D. R. Brill, R. W. Lindquist, Phys. Rev. 131, 471 (1963).
- R. L. Bowers, E. P. T. Liang, ApJ 188, 657 (1974).
- R. Beig, Phys. Lett. A 69, 153 (1978).
- L. Herrera, N. O. Santos, Phys. Rep. 286, 53 (1997).
- B. V. Ivanov, Phys. Rev. D 65, 104011 (2002).
- M. Bañados, P. G. Ferreira, Phys. Rev. Lett. 105, 011101 (2010).
- P. Pani, V. Cardoso, T. Delsate, Phys. Rev. Lett. 107, 031101 (2011).
- G. J. Olmo, Int. J. Mod. Phys. D 20, 413 (2011).
- T. Jacobson, Class. Quantum Grav. 24, 5717 (2007).
- H. Andréasson, J. Differential Equations 245, 2243 (2008).
- P. Karageorgis, J. G. Stalker, Class. Quantum Grav. 25, 195021 (2008).
- J. S. Read, B. D. Lackey, B. J. Owen, J. L. Friedman, Phys. Rev. D 79, 124032 (2009).
- D. Choudhury et al., Astrophys. J. Lett. 971, L20 (2024).
- T. Salmi et al., Astrophys. J. 974, 294 (2024).
- A. J. Dittmann et al., Astrophys. J. 974, 295 (2024).

Added in version 2.34.1:

- H. A. Buchdahl, Astrophys. J. 140, 1512 (1964), doi:10.1086/148055.
- M. S. R. Delgaty, K. Lake, Comput. Phys. Commun. 115, 395 (1998), arXiv:gr-qc/9809013.
- T. W. Baumgarte, N. Ó Murchadha, H. P. Pfeiffer, Phys. Rev. D 75, 044009 (2007).
- P. H. Puttasiddappa, N. Bretón, S. E. Perez Bergliaffa, Phys. Rev. D 114, 084009 (2026), arXiv:2608.10991.
- A. Alho, J. Natário, P. Pani, G. Raposo, Phys. Rev. D 106, L041502 (2022).
- S. Koranda, N. Stergioulas, J. L. Friedman, Astrophys. J. 488, 799 (1997).
- J. M. Lattimer, M. Prakash, Astrophys. J. 550, 426 (2001).
- C. Breu, L. Rezzolla, Mon. Not. R. Astron. Soc. 459, 646 (2016).
- P. Podsiadlowski et al., Mon. Not. R. Astron. Soc. 361, 1243 (2005).

Added in version 2.34.2:

- D. L. Lee, A. P. Lightman, W.-T. Ni, Phys. Rev. D 10, 1685 (1974).
- C. M. Will, Astrophys. J. 204, 224 (1976).
- C. M. Will, Living Rev. Relativ. 17, 4 (2014).
- P. C. C. Freire, N. Wex, Living Rev. Relativ. 27, 5 (2024).
- H. Hernández, L. A. Núñez, Can. J. Phys. 82, 29 (2004).
- J. M. Lattimer, M. Prakash, Phys. Rep. 621, 127 (2016).
- M. C. Miller et al., Astrophys. J. Lett. 1000, L48 (2026).